

Root [0]

Technical Sheet Only!
Please refer to the subsequent sheets for schematic

Power



File: power.kicad_sch

FPGA_Power



File: fpga_power.kicad_sch

FPGA_Config



File: fpga_config.kicad_sch

JTAG



File: jtag.kicad_sch

Serial



File: serial.kicad_sch

USB



File: usb.kicad_sch

Connectivity



File: connectivity.kicad_sch

Input



File: voltage_input.kicad_sch

Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
Czech Technical University in Prague

Sheet: /
File: Controller_Board.kicad_sch

Title: Root

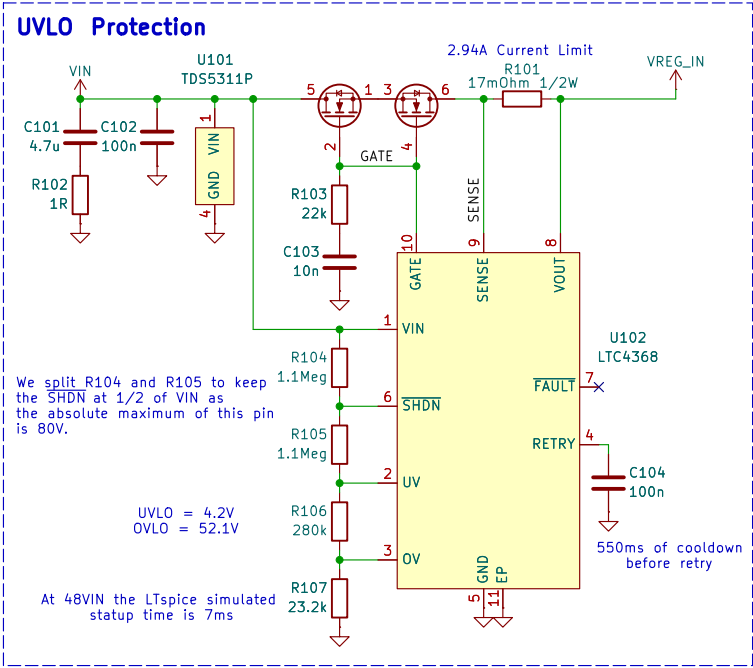
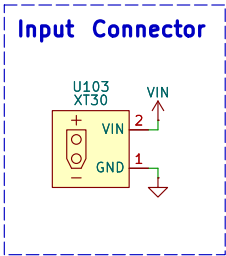
Size: A4 Date: 2026-03-10

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Rev: 1.0

Id: 0/9

Input [1]



$$\text{In}[2] := R3 = \frac{3 \times 10^{-3}}{10 \times 10^{-9}} \times \left(\frac{4 - 0.5}{0.5} \right)$$

$$\text{Out}[2] = 2.1 \times 10^6$$

$$\text{In}[3] := R1 = \frac{\left(\frac{3 \times 10^{-3}}{10 \times 10^{-9}} \right) + 2.1 \times 10^6}{52} \times 0.5$$

$$\text{Out}[3] = 23076.9$$

$$\text{In}[4] := R2 = \left(\frac{3 \times 10^{-3}}{10 \times 10^{-9}} \right) - 23076.9$$

$$\text{Out}[4] = 276923.$$

$$\text{In}[14] := \text{NSolve} \left[2.2 \times 10^6 = \frac{3 \times 10^{-3}}{10 \times 10^{-9}} \times \left(\frac{\text{UV} - 0.5}{0.5} \right), \text{UV} \right]$$

$$\text{Out}[14] = \{ \{ \text{UV} \rightarrow 4.16667 \} \}$$

$$\text{In}[15] := \text{NSolve} \left[24000 = \frac{\left(\frac{3 \times 10^{-3}}{10 \times 10^{-9}} \right) + 2.2 \times 10^6}{\text{OV}} \times 0.5, \text{OV} \right]$$

$$\text{Out}[15] = \{ \{ \text{OV} \rightarrow 52.0833 \} \}$$

UV/OV Calculation

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Sheet: /Input/
File: voltage_input.kicad_sch

Title: Input

Size: A4 Date: 2026-03-27

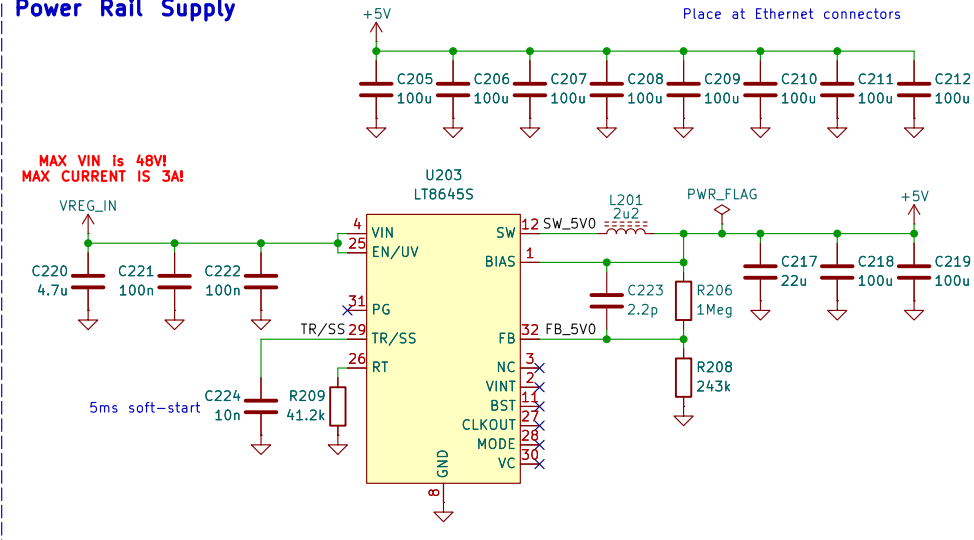
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Rev: 1.0

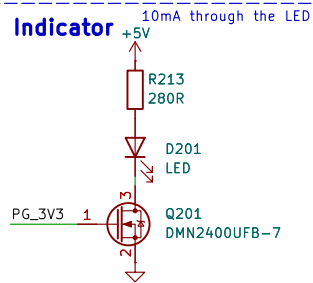
Id: 1/9

Power [2]

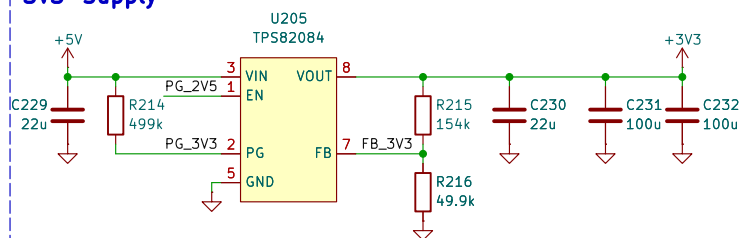
Power Rail Supply



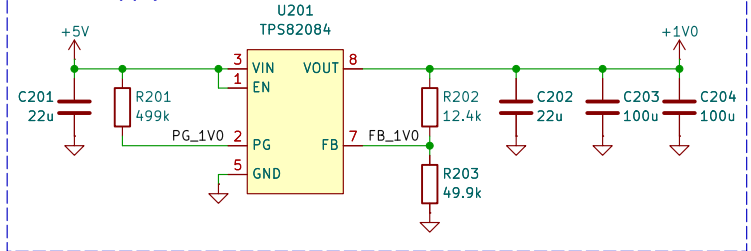
PG Indicator



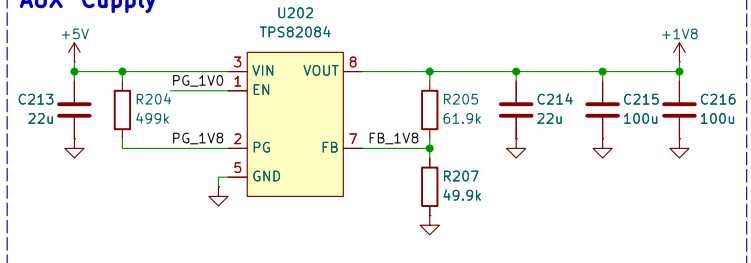
3V3 Supply



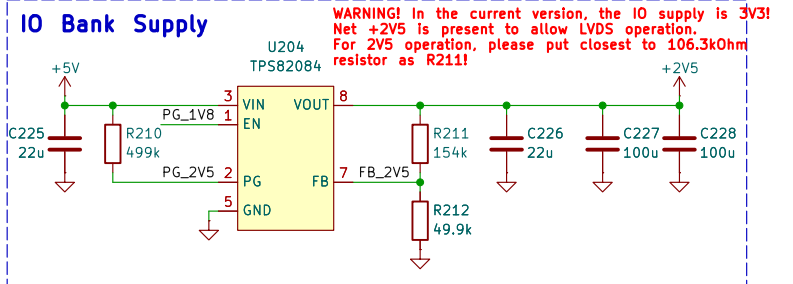
Core Supply



AUX Supply



IO Bank Supply



$$\ln[1] = \text{NSolve}\left[1 = 0.8 \cdot \left(1 + \frac{R1}{50 \cdot 10^3}\right)\right]$$

$$\text{Out}[1] = \{\{R1 \rightarrow 12\,500.\}\}$$

$$\ln[3] = \text{NSolve}\left[1.8 = 0.8 \cdot \left(1 + \frac{R1}{50 \cdot 10^3}\right)\right]$$

$$\text{Out}[3] = \{\{R1 \rightarrow 62\,500.\}\}$$

$$\ln[4] = \text{NSolve}\left[2.5 = 0.8 \cdot \left(1 + \frac{R1}{50 \cdot 10^3}\right)\right]$$

$$\text{Out}[4] = \{\{R1 \rightarrow 106\,250.\}\}$$

$$\ln[2] = \text{NSolve}\left[3.3 = 0.8 \cdot \left(1 + \frac{R1}{50 \cdot 10^3}\right)\right]$$

$$\text{Out}[2] = \{\{R1 \rightarrow 156\,250.\}\}$$

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Sheet: /Power/
File: power.kicad_sch

Title: Power

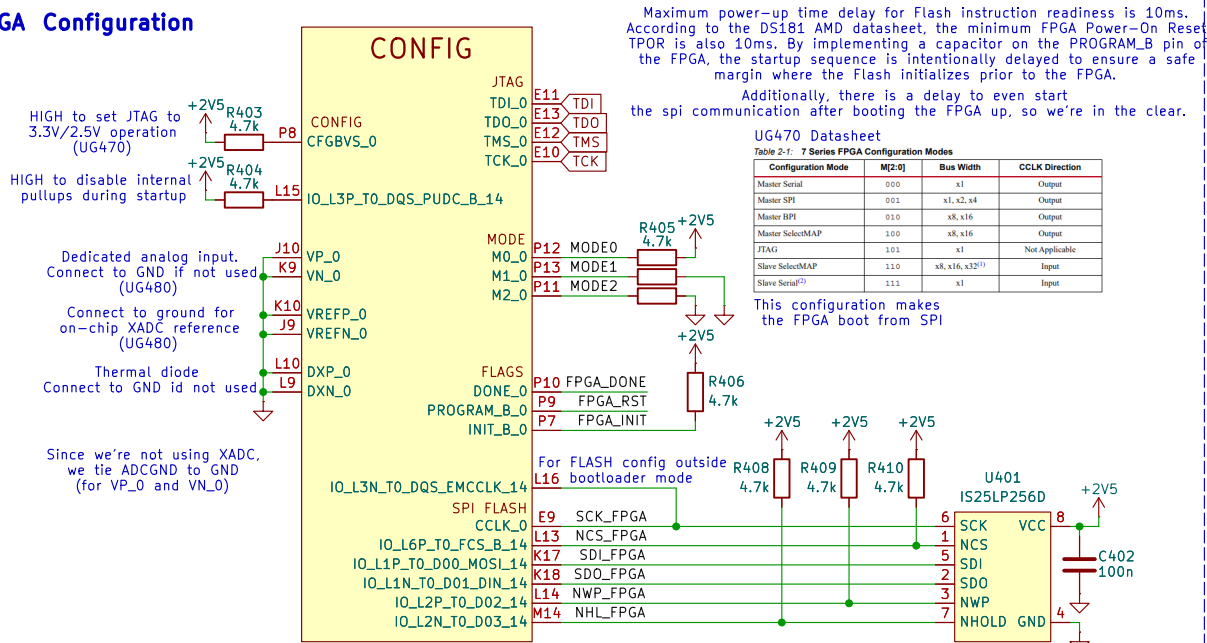
Size: A4
Date: 2026-03-07
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Rev: 1.0
Id: 2/9

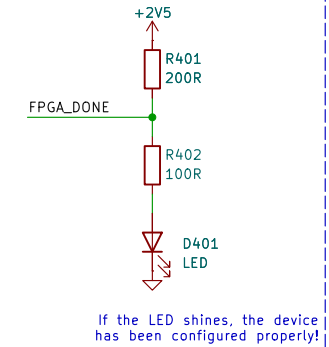
WARNING! BANK 35 is ALWAYS connected to +3V3!
Do not plug LVDS devices into BANK 35 ports!

FPGA Config [4]

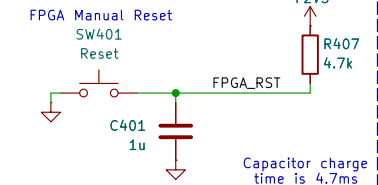
FPGA Configuration



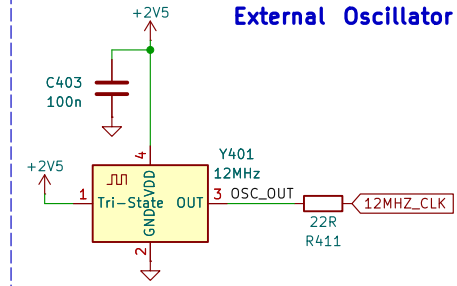
Configuration DONE



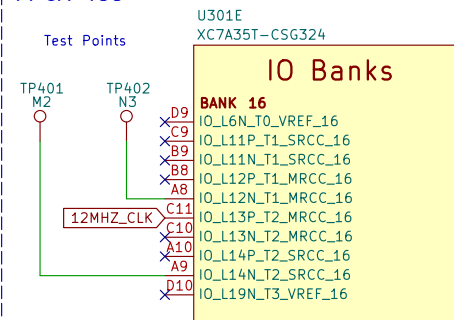
FPGA Manual Reset



External Oscillator



FPGA IOs



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Sheet: /FPGA_Config/
File: fpga_config.kicad_sch

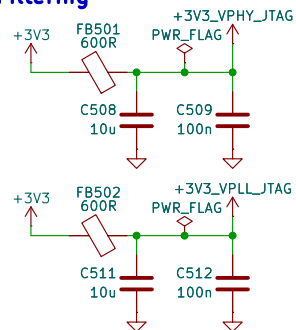
Title: FPGA Config

Size: A4
Date: 2026-03-10
KiCad E.D.A. 10.0.0

Rev: 1.0
Id: 4/9

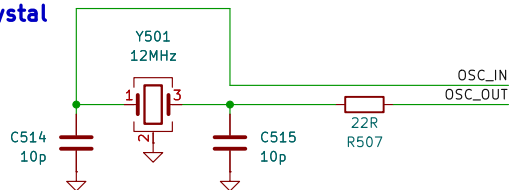
JTAG [5]

Filtering



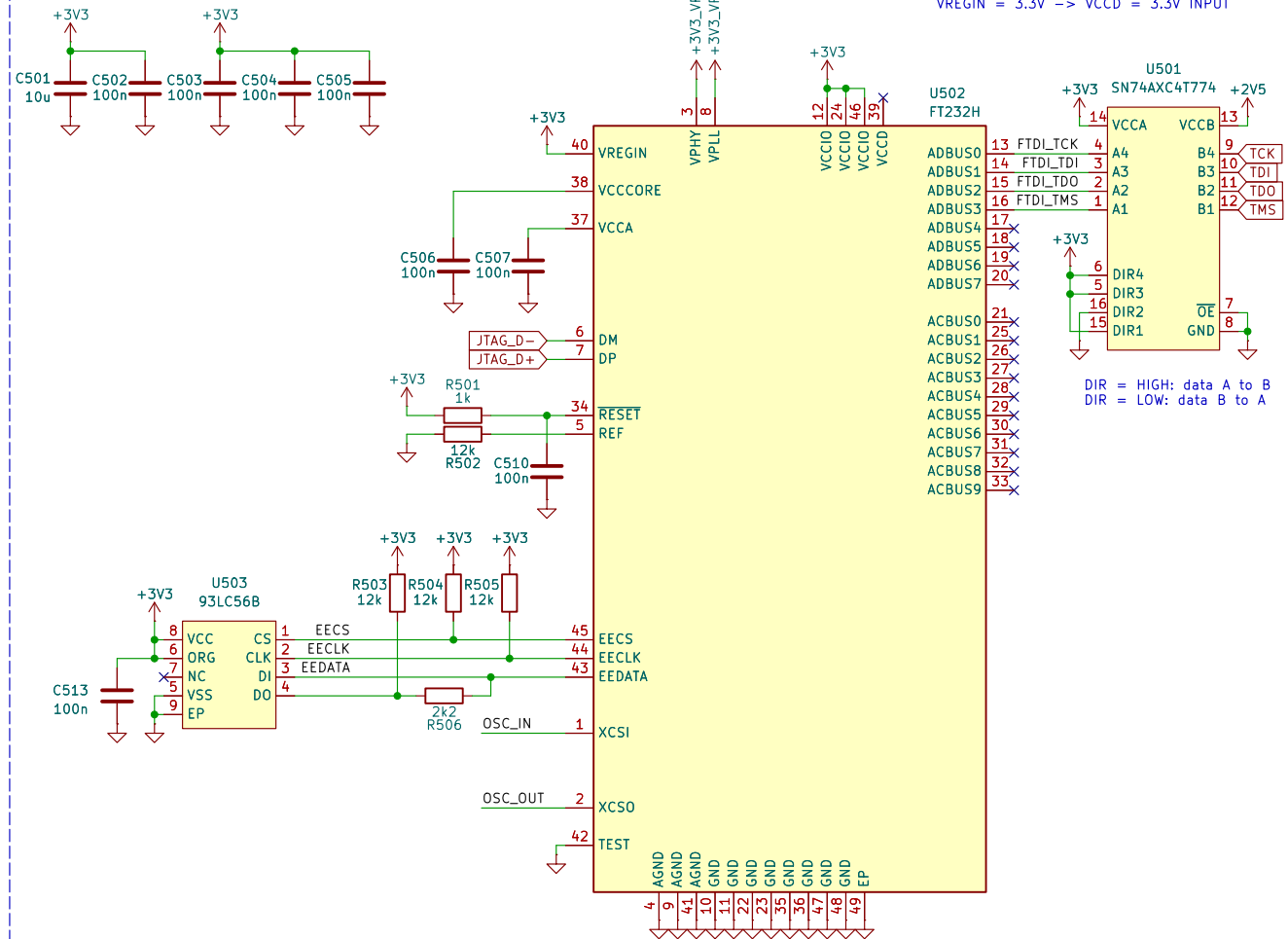
Ferrite beads must be rated at least for 600mA

Crystal



$(10\text{pF Load Capacitance} - 5\text{pF Stray Capacitance}) * 2 = 10\text{pF Load Capacitors}$

FTDI USB-to-JTAG



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Sheet: /JTAG/

File: jtag.kicad_sch

Title: JTAG

Size: A4 Date: 2026-03-11

KiCad E.D.A. 10.0.0

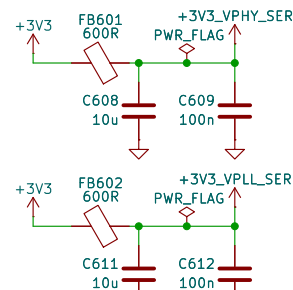
Rev: 1.0

Id: 5/9

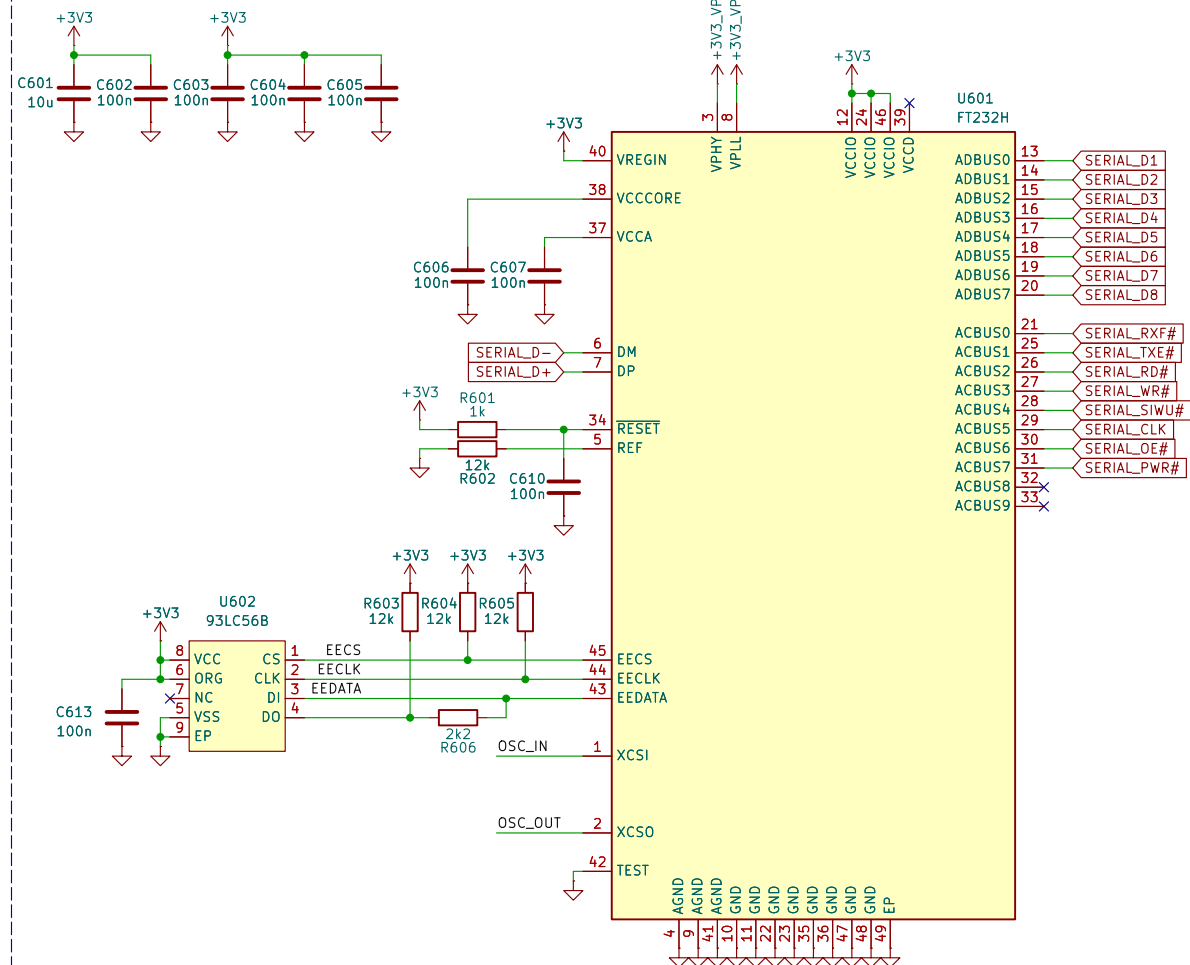
Serial [6]

FTDI USB-to-UART/SYNC FIFO

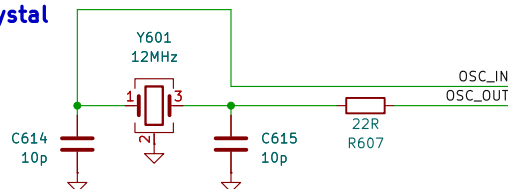
Filtering



Ferrite beads must be rated at least for 600mA



Crystal



$(10\text{pF Load Capacitance} - 5\text{pF Stray Capacitance}) * 2 = 10\text{pF Load Capacitors}$

Reviewer: Michal Varsanyi
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Sheet: /Serial/
File: serial.kicad_sch

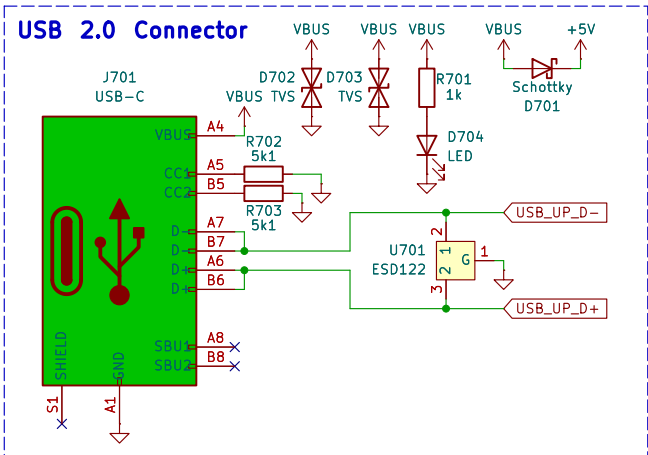
Title: Serial

Size: A4
Date: 2026-03-11
KiCad E.D.A. 10.0.0

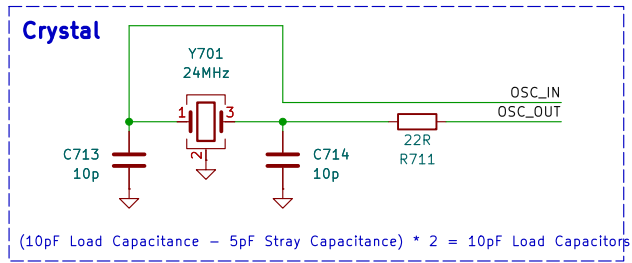
Rev: 1.0
Id: 6/9

USB [7]

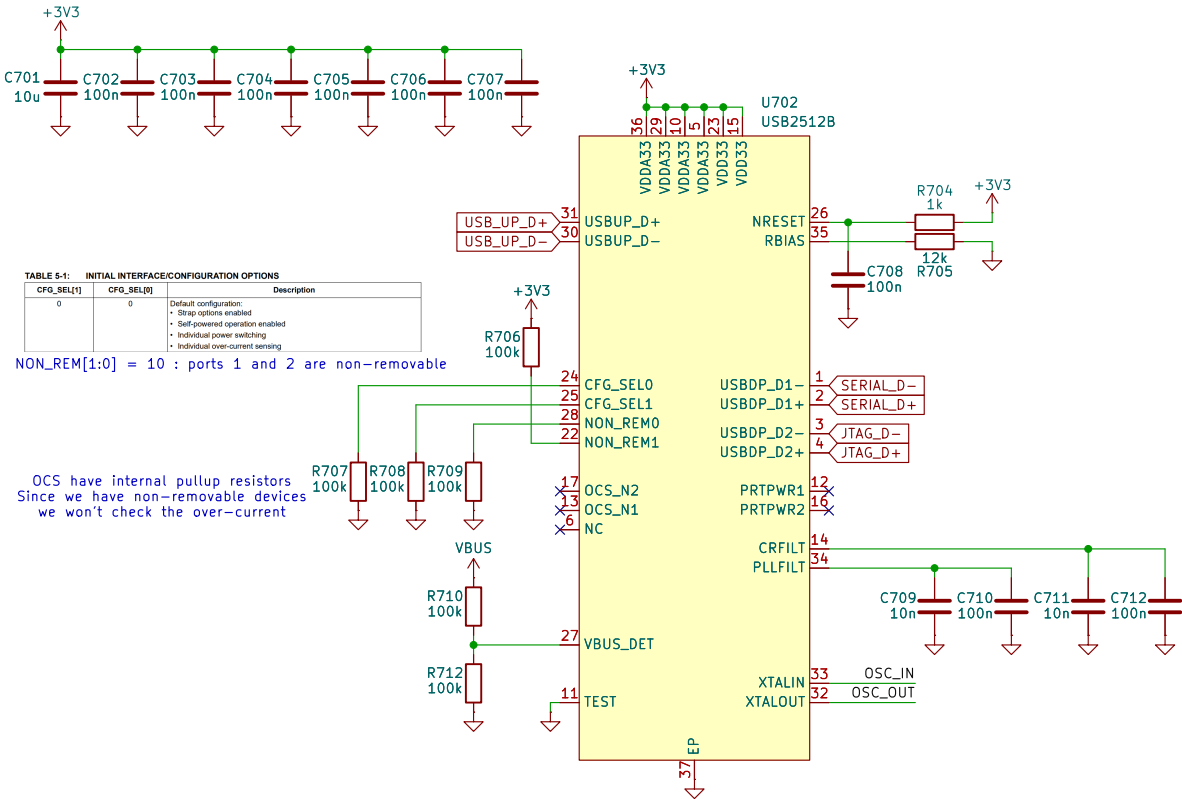
USB 2.0 Connector



Crystal



USB 2.0 Hub



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Sheet: /USB/
File: usb.kicad_sch

Title:

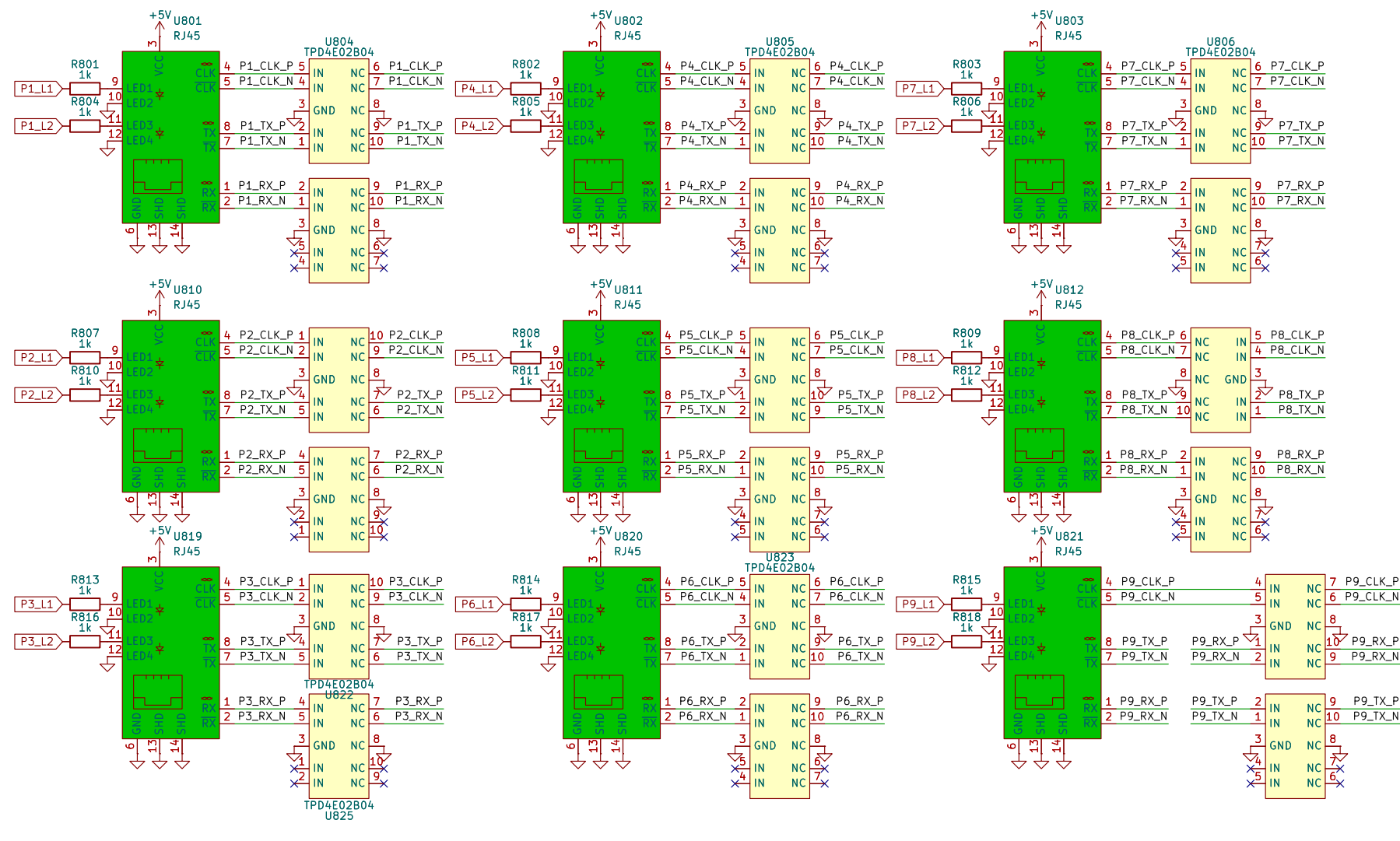
Size: A4
KiCad E.D.A. 10.0.0

Date:

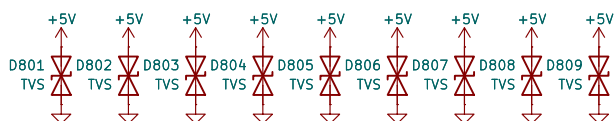
Rev: 1.0
Id: 7/9

Connectivity [8]

Ethernet Connectors



ESD Protection

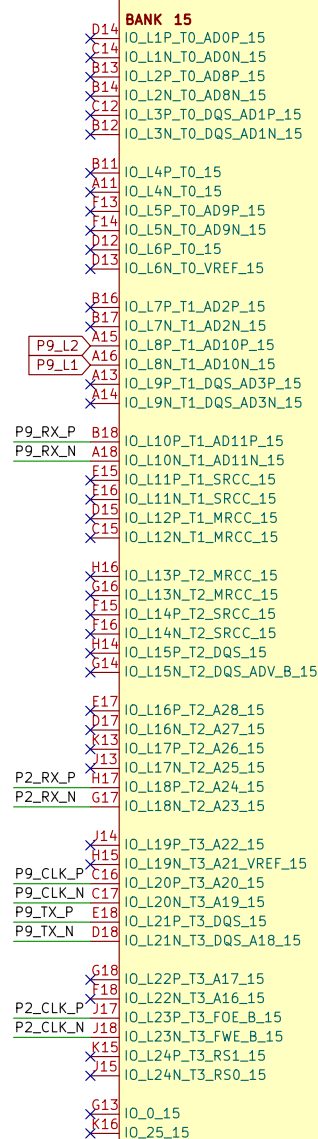


M3 Mounting Holes

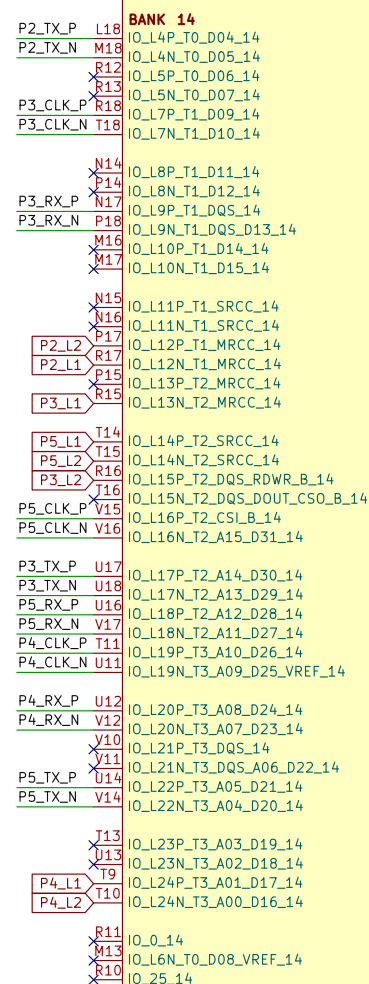


Please excuse this mess, I'll try to make this pretty later

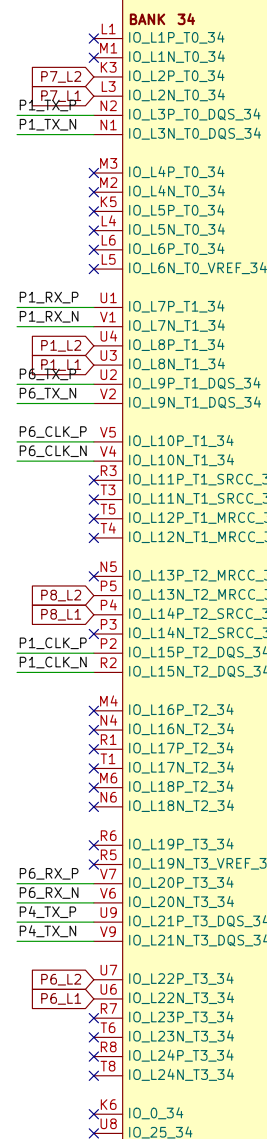
IO Banks



IO Banks



IO Banks



Reviewer: Michal Varsanyi
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Sheet: /Connectivity/
File: connectivity.kicad_sch

Title: Connectivity

Size: A3 Date: 2026-03-12
KiCad E.D.A. 10.0.0

Rev: 1.0
Id: 8/9